

Unsupervised Learning and Clustering: Theory

1. Introduction to Unsupervised Learning and Clustering

1.1 What is Unsupervised Learning?

Unsupervised learning is a branch of Machine Learning in which algorithms are trained on data that has no predefined labels or target outcomes. Instead of learning a mapping from input to a known output, the algorithm's task is to discover the underlying structure, patterns, or groupings that exist naturally within the data. Clustering, dimensionality reduction, and association-rule mining are all examples of unsupervised techniques. The success of an unsupervised model is judged using internal or structural criteria — for example, how compact and well-separated the discovered groups are — rather than a prediction-accuracy metric.

1.2 How Does it Differ from Regression and Classification?

Regression and classification are both supervised learning tasks: the training data includes an input (features) paired with a known output (a continuous value for regression, or a discrete class label for classification). The model learns a function that maps inputs to these known outputs, and its performance can be measured directly by comparing predictions to the true values (e.g., using RMSE, accuracy, or F1-score).

Unsupervised learning, in contrast, works only with the input features; there is no target variable to predict. Clustering algorithms do not know in advance which group a data point belongs to — they infer group membership purely from similarity or distance between points. This makes unsupervised learning exploratory in nature: it is used to understand data, generate hypotheses, or create new features (such as a cluster label) rather than to make a specific prediction.

1.3 Real-Life Examples

Clustering example: A retail company analyses its customers' purchase histories without any predefined categories and discovers that customers naturally group into segments such as "budget-conscious frequent shoppers", "high-spending occasional buyers", and "seasonal shoppers". No labels existed beforehand — the groups emerge from the data itself.

Supervised learning example: A bank builds a credit-scoring model using historical loan data where each past applicant is labelled as either "defaulted" or "repaid". The model learns from these known labels to classify new applicants, and its accuracy can be verified against real outcomes.

2. Clustering Algorithms

2.1 K-Means

Basic idea: K-Means partitions the data into a pre-specified number of clusters, k . It starts by placing k centroids (randomly or via a smarter initialisation such as k-means++), assigns every data point to its nearest centroid using Euclidean distance, and then recomputes each centroid as the mean of the points assigned to it. This assignment-and-update cycle repeats until the centroids stabilise (convergence).

Real-world use case: Segmenting customers of an e-commerce store by annual spending across product categories, so that marketing campaigns can be tailored to each spending profile.

Advantages: Computationally efficient and easy to scale to large datasets; simple to interpret; works well when clusters are roughly spherical and similarly sized.

Limitations: Requires k to be chosen in advance; sensitive to the initial placement of centroids and to outliers; struggles with clusters of irregular shape or very different densities, since it relies on Euclidean distance to a single centroid.

2.2 Hierarchical Clustering

Basic idea: Hierarchical clustering builds a tree of clusters (a dendrogram) rather than a single fixed partition. In the common agglomerative (bottom-up) variant, every point starts as its own cluster; the two closest clusters are merged repeatedly, according to a chosen linkage criterion until all points belong to one cluster. The dendrogram can then be "cut" at any height to obtain a chosen number of clusters.

Real-world use case: Grouping genes with similar expression patterns in bioinformatics, or organising documents into a topic taxonomy where nested categories are meaningful.

Advantages: Does not require the number of clusters to be specified beforehand; the dendrogram gives a rich, multi-level view of how clusters relate to one another; deterministic (no random initialisation).

Limitations: Computationally expensive for large datasets (typically $O(n^2)$ or worse in memory and time); once a merge is made it cannot be undone, so early mistakes propagate; sensitive to noise and to the choice of linkage method.

2.3 DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

Basic idea: DBSCAN groups together points that are densely packed, marking points in low-density regions as noise/outliers. It uses two parameters: ϵ (the neighbourhood radius) and min_samples (the minimum number of points required within that radius for a point to be considered a "core point"). Clusters are formed by connecting core points that lie within ϵ of one another; points that cannot be connected to any dense region are labelled as noise.

Real-world use case: Identifying geographic hotspots of activity — for example, detecting dense areas of crime incidents on a map, or spotting fraudulent transaction clusters, while automatically flagging isolated, unusual events as outliers.

Advantages: Does not require the number of clusters to be specified; can discover clusters of arbitrary shape (not just spherical); naturally identifies and separates out noise/outliers.

Limitations: Performance depends heavily on the choice of ϵ and min_samples , which can be difficult to tune; struggles when clusters have very different densities; performs poorly in high-dimensional feature spaces where distance measures become less meaningful.

2.4 Comparative Summary

K-Means is the fastest and simplest of the three, but it forces every point into a cluster and assumes convex, similarly sized groups. Hierarchical clustering trades speed for a flexible, multi-resolution view of the data without needing k in advance, but does not scale well. DBSCAN uniquely handles irregularly shaped clusters and noisy data, making it well suited to spatial or anomaly-detection problems, but it is sensitive to parameter tuning and less effective in high dimensions. In practice, the choice of algorithm depends on the expected shape of the clusters, the size of the dataset, and whether the presence of outliers must be explicitly modelled.

3. Clustering Metrics

Because unsupervised learning has no ground-truth labels, cluster quality must be assessed with internal validation metrics that examine the geometry of the clusters themselves.

3.1 Elbow Method (SSE)

The Elbow Method plots the Sum of Squared Errors (SSE, also called inertia) — the total squared distance between each point and its assigned cluster centroid — against different values of k . As k increases, SSE always decreases, because more centroids can fit the data more closely. The method looks for the point on the curve where the rate of decrease sharply slows, forming a visual "elbow"; this point is taken as a reasonable trade-off between model complexity and fit.

3.2 Silhouette Score

The Silhouette Score measures, for each point, how similar it is to points in its own cluster (cohesion) compared with points in the nearest other cluster (separation). It is calculated as $(b - a) / \max(a, b)$, where a is the average distance to points in the same cluster and b is the average distance to points in the nearest neighbouring cluster. The overall silhouette score is the mean of this value across all points, ranging from -1 to $+1$, with higher values indicating clusters that are both dense and well separated.

3.3 Davies–Bouldin Index

The Davies–Bouldin Index (DBI) evaluates clustering quality by comparing, for every pair of clusters, the sum of their internal scatter (average distance of points to their centroid) relative to the distance between the two centroids. The index is the average of the worst (highest) such ratio for each cluster. Because it rewards compact clusters that are far apart, lower DBI values indicate better clustering, with no theoretical upper limit.

3.4 Comparison Table

Metric	What it Measures	How it Behaves	What a Good Value Looks Like
Elbow Method (SSE / Inertia)	The within-cluster Sum of Squared Errors — total distance of each point from its cluster centroid.	Decreases monotonically as k increases; the rate of decrease slows after the optimal k .	A visible 'elbow' or bend in the curve, after which adding more clusters gives only marginal SSE reduction.
Silhouette Score	How well each point fits its own cluster compared to the nearest neighbouring cluster (cohesion vs. separation).	Computed per point and averaged; ranges from -1 to $+1$.	Close to $+1$ (well separated, compact clusters). Values near 0 indicate overlapping clusters; negative values indicate misassignment.
Davies–Bouldin Index (DBI)	The ratio of within-cluster scatter to between-cluster separation, averaged over all cluster pairs.	No fixed upper bound; lower values indicate better-defined clusters.	Values close to 0 (dense, well-separated clusters). Higher values indicate overlapping or poorly separated clusters.

4. Choosing k and Interpreting Segments

4.1 How to Choose the Number of Clusters for K-Means

Choosing k is typically done by combining several approaches rather than relying on one metric alone. The Elbow Method offers a quick visual estimate of where additional clusters stop providing meaningful reductions in SSE. The Silhouette Score can be computed for a range of k values, with the k that maximises the average silhouette score generally preferred, since it indicates well-separated and cohesive clusters. The Davies–Bouldin Index can be used similarly, favouring the k that minimises the index. Beyond these statistical criteria, domain knowledge and interpretability matter: a mathematically optimal k is not useful if the resulting segments cannot be meaningfully explained or acted upon by the business. In practice, analysts often compare two or three candidate values of k and select the one that gives both good metric scores and clusters that make practical sense.

4.2 Interpreting Spending Patterns in the Wholesale Distributor Project

In the classic wholesale-customer dataset, each customer's annual spending is recorded across categories such as Fresh, Milk, Grocery, Frozen, Detergents_Paper, and Delicatessen. A cluster with high spending in Fresh and Milk but low spending in Grocery and Detergents_Paper typically represents businesses that need perishable, day-to-day supplies in large quantities that prepare and serve fresh food daily and have limited storage capacity for packaged goods.

Conversely, a cluster with high spending in Grocery and Detergents_Paper but comparatively low Fresh and Milk spending is more indicative of retail-type businesses, such as supermarkets or convenience stores, which stock large volumes of packaged, non-perishable, and cleaning products intended for resale to end consumers, and which typically source fresh perishables through different, more specialised supply chains. Identifying this distinction allows the distributor to tailor delivery schedules (e.g., more frequent, smaller deliveries for HoReCa clients) and sales strategies (e.g., bulk discounts on packaged goods for retail clients) to each segment.

4.3 Why Exclude Channel and Region from the Clustering Features

Channel and Region are categorical variables that describe how a customer is already classified (e.g., as a Retail or HoReCa channel) or where they are located, rather than what they actually purchase. Including them directly in the clustering would bias the algorithm toward regrouping customers according to a label that is already known, rather than allowing the natural, unlabelled structure of purchasing behaviour to emerge from the six spending features. Excluding them keeps the clustering purely behaviour-driven; afterwards, Channel and Region can be reintroduced as external variables to validate whether the discovered clusters align with (or diverge from) these known categories, which is a valuable sanity check but should not influence the clustering itself.

5. Real-World Case Study

5.1 Overview

A relevant real-world example is the study "An Exploration of Clustering Algorithms for Customer Segmentation" (published in *Businesses*, MDPI, 2024), which applies unsupervised learning to a large UK-based online retail transaction dataset obtained from the UCI Machine Learning Repository.

5.2 Goal

The goal of the study was to help retail firms move beyond simple, rule-based customer classification and instead develop a data-driven segmentation model that could support more accurate and efficient marketing and customer-relationship decisions.

5.3 Data Used

The researchers used a transactional dataset containing 541,909 records and eight features, including invoice number, product (stock) code, product description, quantity purchased, invoice date, unit price, customer ID, and country. Rather than clustering on raw transactions, the authors first engineered customer-level features using the RFM framework — Recency (how recently a customer purchased), Frequency (how often they purchase), and Monetary value (how much they spend) — condensing thousands of transactions per customer into three interpretable behavioural metrics.

5.4 Clustering Method Applied

The study compared several clustering algorithms on the RFM-transformed data, including K-Means, Gaussian Mixture Models, DBSCAN, Agglomerative (hierarchical) clustering, and BIRCH. The optimal number of clusters for the partition-based methods was determined using the Elbow Method on inertia values, evaluated across a range of candidate k values.

5.5 Key Results and Insights

The RFM-based clustering successfully separated customers into distinct behavioural segments — for instance, distinguishing frequent, high-monetary, recently active customers (high-value loyal customers) from infrequent, low-spending, long-dormant customers (at-risk or lapsed customers). Comparing algorithms showed that different methods captured different aspects of the customer base: partition-based methods such as K-Means produced clean, easily interpretable segments suited to marketing activation, while density-based approaches were more effective at isolating unusual or outlying purchasing behaviour. The key business insight was that combining a simple, interpretable behavioural model (RFM) with clustering allowed the retailer to prioritise retention efforts on high-value segments and design targeted reactivation campaigns for at-risk segments, illustrating how unsupervised learning can convert raw transactional data into actionable marketing strategy.

References

An Exploration of Clustering Algorithms for Customer Segmentation. *Businesses*, MDPI. Retrieved from <https://www.mdpi.com/2813-2203/2/4/42>

UCI Machine Learning Repository — Wholesale customers dataset and Online Retail dataset.

Course materials and lecture notes on unsupervised learning and clustering algorithms (K-Means, Hierarchical Clustering, DBSCAN).